

The empirical recurrence for OEIS A253349, proved by transfer matrix

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Abstract

OEIS A253349 counts the arrays over $\{0, \dots, 1\}$ whose 2×2 subblocks carry a statistic that is monotone along named directions of the subblock grid, and carries an empirical recurrence of order 84 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. Every direction named moves the subblock row by at most one, so a pair of consecutive array rows is a state, the count is a walk count on $S = 6740$ of them, and the conjectured recurrence is then settled by a finite exact computation.

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1 The conjecture

OEIS A253349 is “Number of $(n+1) \times (7+1)$ 0..1 arrays with every 2×2 subblock diagonal maximum minus antidiagonal maximum nondecreasing horizontally, vertically and ne-to-sw antidiagonally.”. It has offset 1 and begins

6740, 263920, 8338384, 294030700, 9895936900, 340810754556, 11626813328444, 398491692466316, ...

The entry states, as a conjecture contributed by its author and never marked settled:

Empirical recurrence of order 84 (see link above)

Empirical: $a(n) = 144*a(n-1) - 8563*a(n-2) + 255626*a(n-3) - 2972992*a(n-4) - 46865696*a(n-5) + 2174714198*a(n-6) - 26363209292*a(n-7) - 112414254625*a(n-8) + 6919525630552*a(n-9) - 65770551096007*a(n-10) - 282984760153574*a(n-11) + 10898425278688666*a(n-12) - 64898443773624280*a(n-13) - 509497988431144684*a(n-14) + 9023085889649185624*a(n-15) - 21958507892168020728*a(n-16) - 455461716972203949952*a(n-17) + 3799039748390986862448*a(n-18) + 4433114703051246288096*a(n-19) - 200052686711572114925600*a(n-20) + 686859194162479673538688*a(n-21) + 4940440258658882785166528*a(n-22) - 42861549636616322814823296*a(n-23) - 6821543682345866122507136*a(n-24) + 1276418427626083020838706176*a(n-25) - 3759259444397634549979154432*a(n-26) - 20433064806078100617166565376*a(n-27) + 140060516364157736689117370368*a(n-28) + 66061311680920187236972216320*a(n-29) - 2824601347579550805634990071808*a(n-30) + 5197275357941475425138111643648*a(n-31) + 34673731532274580141302886629376*a(n-32) - 146700500271443643421781858713600*a(n-33) - 202436954331656467469817059868672*a(n-34) + 2227482921247546993592526648115200*a(n-35) - 1332442470985324075514993316462592*a(n-36) - 22188739897447961765003590490914816*a(n-37) + 46394914857177354946542084701552640*a(n-38) + 142397942155554769243849133370900480*a(n-39) - 589903106624180351845014444171067392*a(n-40) - 406674249560390571307853683338772480*a(n-41) + 4863900259781355516783542223388016640*a(n-42) - 2597009522419400327422478690530885632*a(n-43) - 28340061646867975912507647836362375168*a(n-44) + 43349972449937961409012505193117384704*a(n-45)$

45) + 115075809285136499484531873747163938816*a(n-46) - 321219473700389402249418929872354410490
 47) - 276240492709476963626791985695871205376*a(n-48) + 16150801061400980769159080124260789780
 49) - 67829317602858236867109071108725276672*a(n-50) - 5983224971355020415874290121688945262592
 51) + 4105967906524877040636631547218701058048*a(n-52) + 1659826651328957015145354532365610359
 53) - 21566054982015635271708471685132241076224*a(n-54) - 3362144365464001348169670024146470096
 55) + 70920845321328809620127974169438970707968*a(n-56) + 44474530781561361116665468898964049
 57) - 170244267338966127203959310238663013564416*a(n-58) - 174155078812951219484756933388697421
 59) + 311989189801922413120110160542652725985280*a(n-60) - 81760927533321136438125758706121793
 61) - 442478918818680196873088206617503189696512*a(n-62) + 24715523706323919903789097336592308
 63) + 484611512535332207381410503923684335419392*a(n-64) - 40775205308449059966316745799752897
 65) - 403257279695238584216799667225660512796672*a(n-66) + 47032994933901802752247337821816470
 67) + 245304908276184523235688932684683162943488*a(n-68) - 40211105268560191820100255909765833
 69) - 98941207073180465130119968605814862643200*a(n-70) + 258619526528163757019026180531273787
 71) + 17279744829170662913558789767350359949312*a(n-72) - 124581956757074656434091250756619866
 73) + 6886996406711814739611146223493274664960*a(n-74) + 4412532173785739259155883059869490400
 75) - 6136172761090018174078184589407457116160*a(n-76) - 11097661645428896952578532590717465788
 77) + 2158647152314394893587982915202470379520*a(n-78) + 1864838958848794254111740130935854596
 79) - 423808058266267022884767335345396121600*a(n-80) - 186516008106577195393117844082203295744
 81) + 44891127045098059416185166577910415360*a(n-82) + 8344086018758729695805334307095969792*a
 83) - 1979459229967184318566328995852320768*a(n-84) for n>87

As of the “Last modified” line on the live entry (Jul 23 2025 13:58:58, revision 6) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

The 2×2 subblocks of an $(n+1) \times 8$ array over $\{0, \dots, 1\}$ form a grid of n rows and 7 columns; write $\sigma(i, j)$ for the value the entry’s statistic takes on the subblock in position (i, j) , a function of its four entries $\begin{pmatrix} p & q \\ r & s \end{pmatrix}$ alone. The condition is

$$\sigma(i, j) \leq \sigma(i, j+1), \quad \sigma(i, j) \leq \sigma(i+1, j), \quad \sigma(i, j) \leq \sigma(i+1, j-1)$$

whenever both positions exist.

Three points of the English are fixed by the entries’ own terms rather than guessed: “ne-to-sw antidiagonally” is the offset $(+1, -1)$ on the subblock grid, since the offset $(+1, +1)$ gives 164 where the corresponding entry publishes 126; “diagonal minus antidiagonal sum” is $(p+s)-(q+r)$; and “ne-sw antidiagonal difference” is $q - r$. Each was checked by counting the small arrays directly and comparing with the entry.

3 The count is a walk count

A pair of consecutive array rows determines one whole row of the subblock grid. Take those pairs as states. The comparisons with offset $(0, \pm 1)$ live inside a single subblock row, so they decide which pairs are admissible at all; the comparisons with offset $(\pm 1, \cdot)$ relate two consecutive subblock rows, so they decide which pair may follow which. A step of the walk appends one array row, turning the pair (r, s) into (s, t) , and settles every comparison between the two subblock rows at once. An $(n+1)$ -row array is therefore a walk of $n-1$ steps and every walk comes from exactly one array, so

$$a(n) = \iota^\top M^{n-1} \tau, \quad \iota = \tau = (1, \dots, 1)^\top,$$

on the $S = 6740$ admissible pairs. In particular a is C -finite. The alignment of the walk index with the entry’s own n was checked against its published terms rather than assumed.

4 The proof

Lemma 1. *Let $q(t) = t^{84} - \sum_i c_i t^{84-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M) \tau$ for the corresponding walk index j , so the recurrence holds from some point on if and only if $\iota^\top M^j q(M) \tau = 0$ for all large j , and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+84} - \sum_i c_i M^{j+84-i}$. For the bound, M^S is an integer combination of $I, M, \dots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M) \tau$ obey the monic recurrence given by the characteristic polynomial of M ; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. $\square$

Theorem 2.

$$a(n) = 144a(n-1) - 8563a(n-2) + 255626a(n-3) - 2972992a(n-4) - 46865696a(n-5) + 2174714198a(n-6) - 2$$

for every $n > 87$.

Proof. The vector $w = q(M)\tau$ was formed by 84 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \dots$, again exactly; those integers vanish from the index corresponding to $n = 87 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. $\square$

5 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 13 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: the digraph is built by reading the entry’s English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on those published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Third, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

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